



Course Description

COURSE NAME	Applied Numerical Analysis
CODE	BSIT 326
UNITS	3
LENGTH OF CLASS	8 weeks
COURSE DESCRIPTION	This course provides comprehensive concepts of numerical methods used in computing to solve mathematical problems. Students will learn techniques for solving nonlinear equations, systems of linear equations, interpolation, numerical differentiation and integration, and optimization. Emphasis will be placed on applying these methods to practical, real-world problems, evaluating their accuracy and efficiency, and developing algorithms for computational implementation. The course also focuses on programming applications of numerical methods, equipping students with hands-on experience in implementing numerical algorithms. By the end of the course, students will have a strong foundation for tackling complex numerical challenges across various IT-related fields.
REQUIRED TEXT	Chapra, S. C., & Canale, R. P. (2021). Numerical methods for engineers (8th ed.). McGraw-Hill Education <u>Supplementary books</u> Kiusalaas, J. (2013). Numerical methods in engineering with Python 3 (3rd ed.) Balagurusamy, E. (1999). Numerical methods.
METHOD OF INSTRUCTION	The course is conducted in a combination modality. Students interact with each other and with the faculty in a classroom setting and in an online learning system. Learning will be facilitated through lecture-discussions, presentations, cooperative learning, and case studies.
SCOPE	Student outcomes are measured through professional individual assignments, discussion postings, comprehensive learning assessments, and class participation.



Summary of Graded Work and Assessments

Graded work and assessments offer students the opportunity to show the degree of mastery for each CLO. The following table shows how assessments and CLOs align (link).

Assignments	Totals	Weight	CLOs
Engagement and Professionalism - Laboratory Class Activities (Rubric)	160	16%	1, 2, 3, 4, 5, 6
Week 1 Discussion	50	5%	1
Week 2 Assignment	100	10%	1, 6
Week 3 Assignment	100	10%	2, 6
Week 4 Assignment	100	10%	3, 6
Week 5 Assignment	100	10%	4, 6
Week 5 Exam	70	7%	1, 2, 3, 4, 6
Week 6 Assignment	100	10%	4, 6
Week 7 Assignment	100	10%	5, 6
Week 8 Discussion	50	5%	1
Week 8 Exam	70	7%	1, 2, 3, 4, 5, 6
Total Points/Percentage	1000 Points	100%	



Course Policies

For Westcliff's course policies, please see the [Course Policies](#) document.

Discussion Requirements

For all discussions, the primary response is due by Thursday at 11:59 p.m. Pacific Time. The primary response must be at least 200 words in length and fully address the topic, demonstrating critical thinking and understanding. Each student must then also post a minimum of two responses to other students in the discussion by Sunday night at 11:59 p.m. Pacific Time. Each peer response must be at least 50 words in length and substantively engage with the other student's original post, continuing the discussion in a professional manner. If at any time information or material is brought in from an outside source or website, it must be properly cited following APA 7th edition guidelines and a full reference must be provided.

Assignment Requirements

Each assignment deliverable is specifically defined in the assignment instructions, such as page length, citations and references, audio or video, presentations, tables, etc. For all written assignments, the required page length does not include the cover or references pages. Refer to the specific requirements as stated in each assignment, and reach out to your instructor for additional information as needed. All graded submissions are due by Sunday at 11:59 p.m. Pacific Time.

All written work must adhere to APA 7th edition academic formatting requirements including core components such as the cover page, page numbers, headings, citations, margins, paragraph indentations, left alignment, double spacing throughout, and the final references using hanging indents

Participation Requirements

Students are required to attend each live class session either in person or virtually as stipulated in the course policies. Participation in the live class session is determined by actively engaging, answering or asking questions, providing comments, interacting in group activities, etc., as required by the instructor. Students who are unable to attend the live in-class or virtual sessions must follow the VCS submission requirements as stated in the Course Policies document.

Writing Center

The Westcliff University Writing Center is dedicated to providing quality support to students and faculty. From assignment review, to in-class workshops, to dissertation support, to publication help, the Writing Center is committed to empowering individuals to use the written language to articulate and disseminate knowledge.



Course Learning Outcomes (CLOs)

Learning outcomes are statements that describe significant and essential scholarship that students have achieved and can reliably demonstrate at the end of the course. Learning outcomes identify what the learner will know and be able to do by the end of a course – the essential and enduring knowledge, abilities (skills), and attitudes (values, dispositions) that constitute the integrated learning needed for successful completion of this course. The learning outcomes for this course summarize what students can expect to learn, and how this course is tied directly to the educational outcomes of the degree.

Course Learning Outcomes (CLOs)	PLOs
1. Apply numerical techniques such as nonlinear equations to solve real world problems effectively.	1, 5, 6
2. Analyze linear systems to identify and implement appropriate solutions.	4, 7
3. Implement different interpolation methods for accurate data approximation.	1, 3, 6
4. Evaluate numerical differentiation and integration techniques in real-world scenarios.	1, 6
5. Utilize numerical optimization techniques to enhance computational models.	4, 6, 7
6. Use a programming language to implement numerical methods and create computational solutions.	1, 2, 6, 7



Detailed Course Outline

The following outline provides important assignment details for this course, unit by unit. Students are responsible for all of the assignments given. Please refer to the Detailed Description of Each Grading Criteria in the syllabus for specific information about each assignment.

Week 1

Assignments to complete this week:

- Reading: Introduction (p. 28-111)
 - Introduction to Numerical Methods
 - Characteristics of Numerical Methods
 - Errors in Numerical Methods

Week 1 Live Class Activities

In this live class activity, students will install and configure Python or Matlab (based on instructor's guideline), and explore common execution errors. The activity aims to build foundational skills in using programming tools for numerical computation and debugging.

Teacher's Instructions and Role

- Setup and Demonstration
- Assist class to install and configure Python or Matlab
- Divide the class into small groups depending on class size.
- Assign simple tasks to groups related to errors in numerical methods.
- Monitor group progress and provide assistance as needed.

Group Instructions

- Install and configure Python or Matlab environment
- Work collaboratively in assigned groups.
- Prepare to present findings and solutions during the sharing session.

Each group will share their findings in live class.

Week 1 Discussion ([Rubric](#))

Discuss the concept of Numerical Methods in your own words. How do they differ from analytical methods? Provide two real-world examples where Numerical Methods are necessary, and explain why analytical solutions may not be feasible in these cases.



Week 2

Assignments to complete this week:

- Reading: Roots of Nonlinear equations (p. 124-230)
 - Bisection Method (p. 128)
 - Newton Raphson Method (p. 152)
 - Secant Method (p. 158)

Week 2 Live Class Activities

In this live class activity, students will implement the Bisection Method to estimate a root for the given function using either Python or MATLAB. The activity aims to reinforce the understanding of root-finding techniques and provide hands-on experience with numerical methods.

Teacher's Instructions and Role

- Briefly introduce the Bisection Method and explain how it works for finding roots of a given function.
- Divide the class into small groups based on class size.
- Assign a function [*like* $f(x) = x * \tan(x) - 1$] to each group to implement the Bisection Method.
- Encourage students to debug their code and adjust parameters like tolerance and iteration limits.
- Monitor group progress and provide assistance as needed.

Group Instructions

- Implement the Bisection Method to find the root of a given function.
- Work collaboratively in assigned groups.
- Collaborate with your group to solve the problem and test different parameters (e.g., tolerance).
- Prepare to present findings and solutions during the sharing session.

Each group will share their findings along with steps in live class.

Week 2 Assignment ([Rubric](#))

Estimate one root of the following nonlinear equations using both manual calculations (pen and paper) and computer programming (MATLAB or Python). Show all the working steps for manual calculations and provide the code implementation for each method.

Newton Raphson Method

$$f(x) = e^{-0.5x}(4 - x) - 2$$

Secant Method

$$f(x) = x^2 + \cos(x) - e^{-x} - 2$$



Week 3

Assignments to complete this week:

- Reading: Solution of linear equations (p. 235- 324)
 - Review of matrix
 - Gauss Elimination Method (p. 256)
 - Gauss Jordan Method (p. 277)
 - Gauss Seidel Method (Iterative) (p. 309)

Week 3 Live Class Activities

In this live class activity, students will implement Gauss Elimination, Gauss Jordan, or Gauss Seidel methods to solve a system of linear equations. The goal is to understand each method's steps and compare manual solutions with Python/Matlab results.

Teacher's Instructions

1. Introduce the three methods and their applications.
2. Divide the class into small groups and assign one method per group.
3. Guide and support students as they work through each step.

Group Instructions

1. Solve the given system using your assigned method step-by-step manually as well programming.
2. Work collaboratively to ensure accuracy.
3. Compare manual solutions with Python/Matlab results.
4. Be ready to explain your steps, findings, and observations to the class.

Each group will present their process, results, and key takeaways during the sharing session.

Week 3 Assignment ([Rubric](#))

Solve the following problems using the specified methods by working them out step-by-step manually (with pen and paper) and showing all intermediate steps clearly. Additionally, write a program (Python or Matlab) to solve each problem using the same methods, and provide an explanation of the programming process and logic.

Gauss Elimination Method

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$



$$-8x_1 + x_2 - 2x_3 = -20$$

Gauss Jordan Method

$$2x + 3y - z = 54$$

$$x + y + z = 6$$

$$-2x + 5y + 2z = -3$$

Gauss Seidel Method

$$-3x + y + 15z = 44$$

$$6x - 2y + z = 5$$

$$5x + 10y + z = 28$$



Week 4

Assignments to complete this week:

- Reading: Interpolation (p. 447- 534)
 - Linear Interpolation
 - Newton Interpolation (p. 497)
 - Lagrange Interpolation (p. 508)
 - Chebyshev Interpolation

Week 4 Live Class Activities

In this live class activity, students will implement the Linear Interpolation Method to estimate values between known data points using either Python or MATLAB. The activity aims to strengthen the students' understanding of interpolation techniques and their application in numerical methods.

Teacher's Instructions and Role

- Briefly explain the concept of Linear Interpolation
- Divide the class into small groups based on class size.
- Assign a task to each group to implement the Linear Interpolation method to estimate values for a given dataset. Provide necessary materials like dataset if needed.
- Monitor group progress and provide assistance as needed.

Group Instructions

- Implement the Linear Interpolation method to estimate values between given data points.
- Work collaboratively in assigned groups.
- Prepare to present findings and solutions during the sharing session.

Each group will share their findings along with steps in live class.

Week 4 Assignment ([Rubric](#))

For this assignment, you are required to gather temperature data from a reliable source (such as a weather app or a local thermometer) at six different time points throughout the day. Using this collected data, apply two interpolation methods of your choice-options include Lagrange Interpolation, Newton Interpolation, or Chebyshev Interpolation to estimate the temperature at a specific time between your recorded points, for example, 5 PM.

For each interpolation method, calculate the estimated temperature, compare the results, and evaluate the accuracy of their predictions. Highlight any performance differences between the methods and analyze their computational efficiency. Additionally, discuss the practical applications of each technique and recommend which method would be more suitable for different types of data or varying accuracy requirements.



Submission Structure:

1. **Dataset:** Include the temperature readings and the time points.
2. **Calculations:** Show your work or include your code.
3. **Results and Analysis:** Present your estimates, compare the outcomes, and provide a detailed discussion.

You may complete the assignment using manual calculations or a programming language of your choice.



Week 5

Assignments to complete this week:

- Reading: Numerical Differentiation and Integration (p.596- 684)
 - Numerical Differentiation (p. 667)
 - Trapezoidal Rule (p. 614)
 - Simpson's Rules (p. 624)
 - Gaussian Integration (p. 651)

Week 5 Assignment ([Rubric](#))

In this assignment, you are required to complete the following tasks:

1. *Differentiation Concepts*: Explain the difference between exact differentiation and numerical differentiation, providing clear examples to illustrate both concepts.
2. *Numerical Integration*: Approximate the integral of the function $f(x) = \frac{e^{-x^2}}{1+x^2}$ over the interval $[0,2]$ using the following methods:
 - Trapezoidal Rule
 - Simpson's Rule
 - Gaussian Integration (with 4 subintervals)

Perform the calculations manually for each method. Then, compare the approximate results with the exact value of the integral (obtained numerically). Calculate the error for each method and analyze the accuracy of each approach. Identify which method provides the best approximation and explain the reasons behind its superior performance.

Week 5 Exam

Exam: This 75-minute exam will evaluate the material covered in weeks 1 through 4 and will take place during a live class under the supervision of a proctor.



Week 6

Assignments to complete this week:

- Reading: Ordinary Differential Equations (p. 711- 792)
 - Euler's Method (p. 722)
 - Runge Kutta Method (p. 741)
 - Shooting Method (p. 795)

Week 6 Live Class Activities

In this live class activity, students will use Euler's Method, Runge-Kutta Method, or Shooting Method.

Teacher's Instructions and Role:

- Briefly explain the concept of Euler's Method, Runge-Kutta Method, or Shooting Method.
- Divide the class into small groups based on class size.
- Provide the necessary instruction if needed.
- Monitor group progress, answering questions and providing assistance as required.

Group Instructions and Sharing:

In this live class activity, students will work in groups, each starting with a different problem related to one of the following methods: Euler's Method, Runge-Kutta Method, or Shooting Method. Each group will have 10 minutes to work on their assigned problem and solve as much as possible. After the time is up, the groups will rotate to the next problem. At each new station, they will review the previous group's work, identify any errors, and continue solving the problem. Once all rotations are complete, each group will present the final solution to the problem they initially started with, explaining their method, any challenges faced, and how the contributions of all groups helped reach the final solution.

Week 6 Assignment ([Rubric](#))

Write a program in Python or MATLAB to solve the following problems. Provide a detailed explanation of the process and compare the results obtained using the different mentioned methods.

- a. Solve the following initial-value problem over the interval from $t = 0$ to 2 where $y(0) = 1$. Display all your results on the same graph. $\frac{dy}{dt} = yt^3 - 1.5y$

(a) Analytically. (b) Euler's method with $h = 0.5$ and 0.25 . (c) Fourth-order RK method with $h = 0.5$.



- b. Solve the following problem over the interval from $x = 0$ to 1 using a step size of 0.25 , where $y(0) = 1$. Display all your results on the same graph. $\frac{dy}{dx} = (1 + 2t)\sqrt{x}$
- (a) Analytically. (b) Euler's method. (c) Fourth-order RK method.



Week 7

Assignments to complete this week:

- Reading: Optimization (p. 350- 420)
 - Golden Section Search Method (p. 361)
 - Newton's Method (p. 370)
 - Gradient Method (p. 380)

Week 7 Live Class Activities

In this live class activity, each group will implement one of the methods: Golden Section Search, Newton's Method or Gradient Method.

Teacher's Instructions and Role

- Briefly explain the concept of Golden Section Search, Newton's Method or Gradient Method.
- Divide the class into small groups based on class size.
- Allow students to choose one of the methods.
- Provide other necessary instructions if needed.
- Monitor group progress and provide assistance as needed.

Group Instructions

- Choose one of the methods: Golden Section Search, Newton's Method or Gradient Method to approach the problem and develop both a manual solution and programming (programming code or pseudocode)
- Work collaboratively in assigned groups.
- Prepare to present findings and solutions during the sharing session.

Each group will share their findings along with steps in live class.

Week 7 Assignment ([Rubric](#))

Analytical solutions to optimization problems are often challenging or even impossible to derive in certain scenarios. Provide an example of an optimization problem in the field of artificial intelligence where numerical methods are essential. Describe how numerical computation can be applied to effectively solve the problem.

Week 8

Assignments to complete this week:

- Reading:
 - Applications of Numerical Computations.
 - Review all topics learned from week 1 to week 7 to prepare for the final Exam.

Week 8 Discussion ([Rubric](#))

Discuss the role of numerical computations in training machine learning models. How are numerical methods, such as matrix operations and gradient descent, applied in machine learning algorithms? What common computational challenges arise, and how can they be addressed?

Week 8 Exam

Exam: This 75-minute exam will evaluate the material covered in weeks 5 through 7 and will take place during a live class under the supervision of a proctor.